Chapter 2

Polynomials

Zeroes of a Polynomial

IP Important Points

A real number k is said to be zero of a polynomial $p(x)$ if $p(k) = 0$

Geometrical meaning of the zeroes of polynomial

x coordinate of the point of intersection of the graph of a polynomial with x-axis is known as zero of the polynomial.

Number of zeroes of a polynomial is equal to number of points of intersection of the graph of polynomial with x-axis.

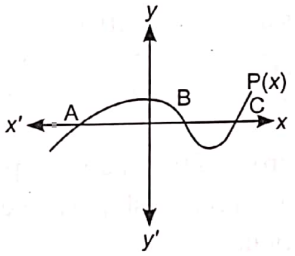
MCQs Multiple Choice Questions (1 Mark)

1. The zeroes of the polynomial $3x^2 + 11x - 4$ are: [2023(C)]

   (a) $\frac{1}{2}, -4$ (b) $\frac{1}{4}, -3$

   (c) $\frac{1}{3}, -4$ (d) $\frac{1}{3}, 4$

2. In figure, the graph of a polynomial P(x) is shown. The number of zeroes of P(x) is [2021]



   (a) 1 (b) 2

   (c) 3 (d) 4

3. The graph of a polynomial P(x) cuts the x-axis at 3 points and touches it at 2 other points. The number of zeroes of P(x) is [2021]

   (a) 1 (b) 2

   (c) 3 (d) 5

4. If $x - 1$ is a factor of the polynomial $p(x) = x^3 + ax^2 + 2b$ and $a + b = 4$, then [2021]

   (a) $a = 5, b = -1$ (b) $a = 9, b = -5$

   (c) $a = 7, b = -3$ (d) $a = 3, b = 1$

5. If one of the zeroes of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is [2020]

   (a) 10 (b) –10

   (c) –7 (d) –2

Relationship Between the Zeroes and Coefficients of a Quadratic Polynomial

IP Important Points

If α and β are zeores of a quadratic polynomial

$p(x) = ax^2 + bx + c$

then

(i) $\alpha + \beta = \frac{-b}{a}$

$\Rightarrow$ sum of zeroes $= \frac{-\text{coefficient of } x}{\text{coefficient of } x^2}$

(ii) $\alpha.\beta = \frac{c}{a}$

$\Rightarrow$ Product of zeroes $= \frac{\text{constant term}}{\text{coefficent of } x^2}$

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If α and β are zeroes of a quadratic polynomial then

$$p(x) = k[x^2 - (\alpha + \beta)x + \alpha\beta]$$

$\Rightarrow$ $p(x) = k[x^2 - (\text{sum of zeroes})x + \text{product of zeroes}]$

where k is any non-zero real number.

[Generally we take $k = 1$]

MCQs Multiple Choice Questions (1 Mark)

6. If α, β are zeroes of the polynomial $x^2 - 1$, then value of $(\alpha + \beta)$ is : [2023]

(a) 2 (b) 1 (c) -1 (d) 0

7. If α, β are the zeroes of the polynomial $p(x) = 4x^2 - 3x - 7$, then $\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$ is equal to : [2023]

(a) $\frac{7}{3}$ (b) $\frac{-7}{3}$

(c) $\frac{3}{7}$ (d) $\frac{-3}{7}$

8. A quadratic polynomial, the product and sum of whose zeroes are 5 and 8 respectively is [2021]

(a) $k[x^2 - 8x + 5]$ (b) $k[x^2 + 8x + 5]$

(c) $k[x^2 - 5x + 8]$ (d) $k[x^2 + 5x + 8]$

9. If α, β are the zeros of the quadratic polynomial $p(x) = x^2 - (k + 6)x + 2(2k - 1)$, then the value of k, if $\alpha + \beta = \frac{1}{2}\alpha\beta$, is [2021]

(a) -7 (b) 7

(c) -3 (d) 3

10. The quadratic polynomial, the sum of whose zeroes is -5 and their product is 6, is [2020]

(a) $x^2 + 5x + 6$ (b) $x^2 - 5x + 6$

(c) $x^2 - 5x - 6$ (d) $-x^2 + 5x + 6$

VSA Very Short Answer Type Questions (2 Marks)

11. If one zero of the polynomial $p(x) = 6x^2 + 37x - (k - 2)$ is reciprocal of the other, then find the value of k. [2023]

12. If the product of zeroes of the polynomial $ax^2 - 6x - 6$ is 4, find the value of a. Find the sum of zeroes of the polynomial. [2014]

13. Find the zeroes of the quadratic polynomial $9t^2 - 6t + 1$ and verify the relationship between the zeroes and the coefficients. [2014]

SA Short Answer Type Questions (3 Marks)

14. Find a quadratic polynomial whose zeroes are reciprocals of the zeroes of the polynomial $f(x) = ax^2 + bx + c$, $a \neq 0$, $c \neq 0$. [2020]

15. Find the zeroes of the quadratic polynomial $7y^2 - \frac{11}{3}y - \frac{2}{3}$ and verify the relationship between the zeroes and the coefficients. [2019]

16. Find the quadratic polynomial, sum and product of whose zeroes are -1 and -20 respectively. Also, find the zeros of the polynomial so obtained. [2019]

17. Find the zeroes of the quadratic polynomial $3x^2 - 2$ and verify the relationship between the zeroes and the coefficients. [2016]

18. Quadratic polynomial $2x^2 - 3x + 1$ has zeroes as α and β. Now form a quadratic polynomial whose zeroes are 3α and 3β. [2016]

19. If α and β are zeroes of a polynomial $x^2 + 6x + 9$, then form a polynomial whose zeroes are $-\alpha$ and $-\beta$. [2014]

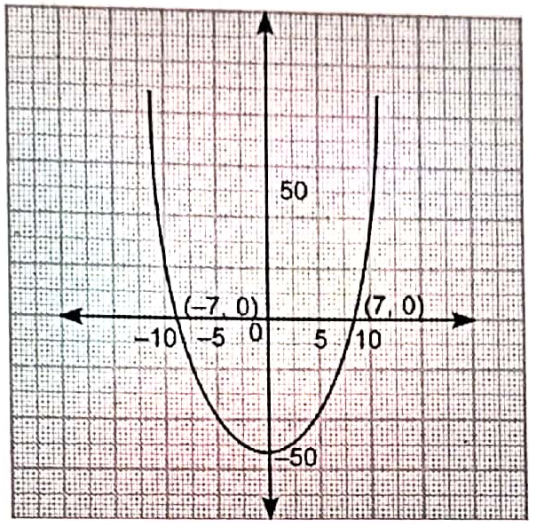
LA Long Answer Type Question (5 Marks)

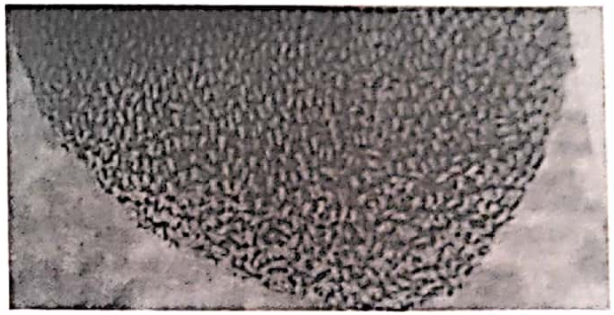
20. If one zero of the quadratic polynomial $f(x) = 4x^2 - 8kx + 8x - 9$ is negative of the other, then find the zeroes of $kx^2 + 3kx + 2$. [2015]

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CBQ Case-Based Question

21. While playing in a garden, Samaira saw a honey-comb and asked her mother what is that. Her mother replied that it's a honeycomb made by honey bees to store honey. Also, she told her that the shape of the honeycomb formed is a mathematical structure. The mathematical representation of the honeycomb is shown in the graph. [2023 (C)]





Based on the above information, answer the following questions:

(*i*) How many zeroes are there for the polynomial represented by the graph given?

(*ii*) Write the zeroes of the polynomial.

(*iii*) If the zeroes of the polynomial $x^2 + (a + 1)x + b$ are 2 and –3, then determine the values of a and b.

OR

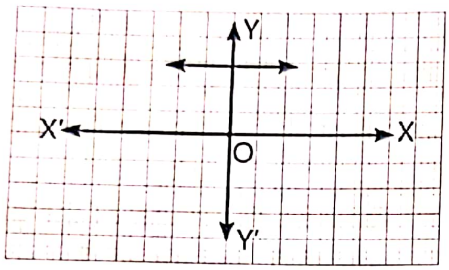
If the square of difference of the zeroes of the polynomial $x^2 + px + 45$ is 144, the find the value of p.

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TRAILBLAZER QUESTIONS

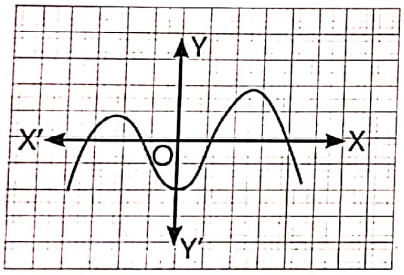
MCQs Type Questions

1. The graphs of $y = p(x)$ is given. then the number zeroes of $p(x)$ is:



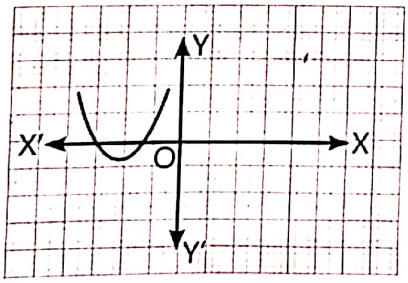
a) 0 b) 1
c) 2 d) 4

2. The graphs of $y = p(x)$ is given. then the number zeroes of $p(x)$ is:



a) 0 b) 1
c) 4 d) 5

3. The graphs of $y = p(x)$ is given. then the number zeroes of $p(x)$ is:



a) 4 b) 2
c) 0 d) 1

4. The Degree of a cubic polynomial is:

a) 0 b) 1
c) 2 d) 3

5. If α and β are zeroes of a quadratic polynom $p(x) = ax^2 + bx + c$ then sum of zeroes can written as

a) $\alpha + \beta = \frac{-(coefficient\ of\ x)}{coefficient\ of\ x^2}$

b) $\alpha + \beta = \frac{(coefficient\ of\ x)}{coefficient\ of\ x^2}$

c) $\alpha + \beta = \frac{-(constant\ term)}{coefficient\ of\ x^2}$

d) $\alpha + \beta = \frac{(constant\ term)}{coefficient\ of\ x^2}$

6. The value of x for which the polynomials -1 and $x^2 - 2x + 1$ vanish simultaneously,

a) 2 b) -2
c) -1 d) 1

7. The quadratic polynomial whose zeroes a -3 and -4 is

a) $x^2 - 7x + 12$
b) $x^2 + 7x + 12$
c) $x^2 + 7x - 12$
d) None of the above

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8. The zeroes of quadratic polynomial $x^2 + 9x + 27$ are
   a) Both positive
   b) Both negative
   c) One positive and one negative
   d) Both equal

9. A ____ is a polynomial of degree 0.
   a) Coefficient
   b) Contant number
   c) None of the above
   d) Both equal

10. (A) Assertion: $x^2 + x$ has only 1 real zero.
(R)Reason: A polynomial of nth degree must have n real zeroes.
   a) Both assertion and reason (R) are true, and reason (R) is the correct explanation of assertion.
   b) Both assertion and reason (R) are true, but reason (R) is not the correct explanation of assertion.
   c) Assertion is false but reason (R) is true.
   d) Assertion is true, but reason (R) is false.

Very short/Short Answer Type Questions

1. Find the zeroes of the following quadratic polynomials $4u^2+8u$ and verify the relationship between the zeroes and the coefficients.

2. Find a quadratic polynomial whose sum of zeroes and product of zeroes are 0, $\sqrt{5}$ respectively.

3. If sum of zeroes of the quadratic polynomial $p(x) = kx^2 + 2x + 3k$ is equal to the product of its zero then find the value of k.

4. Find the zeroes of $p(x) = 2x^2 - x - 6$ and verify the relationship of zeroes with these co-efficient.

5. Find the value k of for which the zeroes of the polynomial $3x^2 - kx + 16$ are in ratio 7:8.

6. If one zero of the quadratic polynomial $f(x) = 4x^2 - 8kx + 9$ is equal to the other, find the value of k.

7. If a and b are zeroes of quadratic polynomial $kx^2 + 4x + 4$, find the value of k such that $(a + b)^2 - 2ab = 24$.

8. If the sum and product of the zeroes of the polynomial $ax^2 - 5x + c$ are equal to 10 each, find the value of a and c.

9. If one of the zeroes of the polynomial $3x^2 - 8x + 2k + 1$ is seven times the other, find the value of k.

10. If α and β are the zeroes of a quadratic polynomial such that $\alpha + \beta = 24$ and $\alpha - \beta = 8$. Find the quadratic polynomial having α and β as its zeroes.

11. If m and n are the zeroes of the quadratic polynomial $f(x) = x^2 - 4x + 3$, find the value of $m^4n^3 + m^3n^4$.

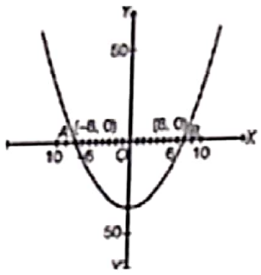
12. If a and b are the zeroes of the polynomial $3x^2 + 11x - 4$, find the value of $\frac{a}{b} + \frac{b}{a}$.

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Case Study Based Questions

1. While playing in garden, Sahiba saw a honeycomb and asked her mother what is that. She replied that it's a honeycomb made by honey bees to store honey. Also, she told her that the shape of the honeycomb formed is parabolic. The mathematical representation of the honeycomb structure is shown in the graph.





i) The Graph of a quadratic polynomial is in _____shape.

a) Straight line b) Parabolic

c) Circular d) None of these

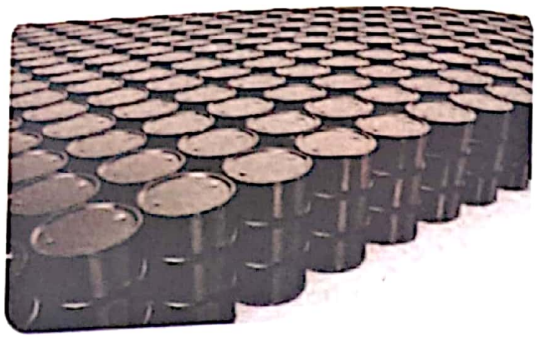
ii) The expression of the polynomial represented by the graph is:

a) x^2-49 b) x^2-64

c) x^2-36 d) x^2-81

iii) Find the value of the polynomial represented by the graph when $x = 6$.

2. A barrels, manufacturer can produce up to 300 barrels per day. The profit made from the sale of these barrels can be modelled by the function $p(x) = -10x^2 + 3500x - 66000$, where $p(x)$ is the profit in rupees and x is the number of barrels made and sold.



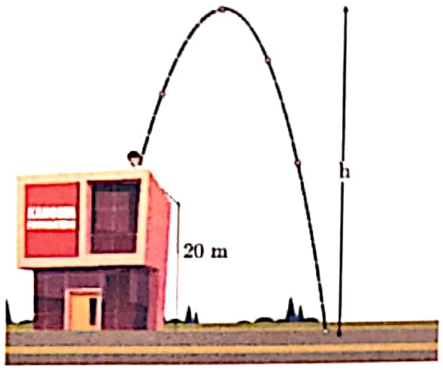
Based on this model, answer the following questions.

i) When no barrels are produced, how much i the profit or loss?

ii) Which of these is the break-even point? (A zero profit point is called the break-eve point.)

iii) What is the profit/loss if 175 barrels are produced?

3. Lavanya throws a ball upwards, from a rooftop, which is 20 m above the ground. It will reach a maximum height and then fall back to the ground. The height of the ball from the ground at time t is h, which is given by $h = -4t^2 + 16t + 20$.



(i) What is the height reached by the ball after 1 second?

(ii) What is the maximum height reached by the ball?

(iii) How long will the ball take to hit the ground?

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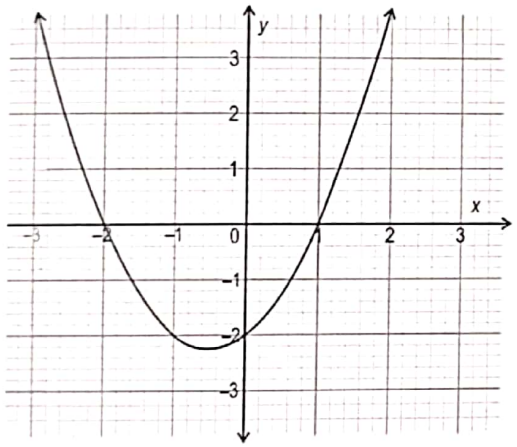
MCQ Type Questions

1. How many zero(es) does $(x-2)(x+3)$ have?
   a) Zero b) One
   c) Two d) Three

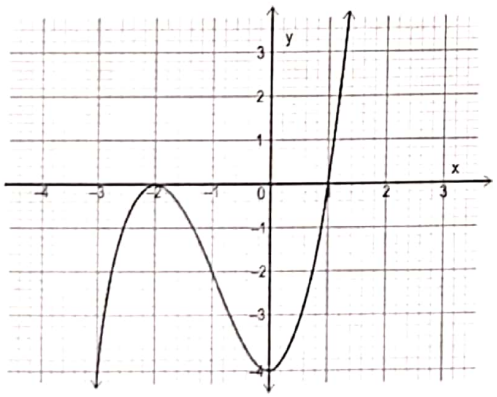
2. How many zero(es) does the polynomial $293x^2 - 293x$ have?
   a) 2 b) 1
   c) 0 d) 3

3. Which of the following could be the graph of the polynomial?
   $(x - 1)2(x + 2)$?

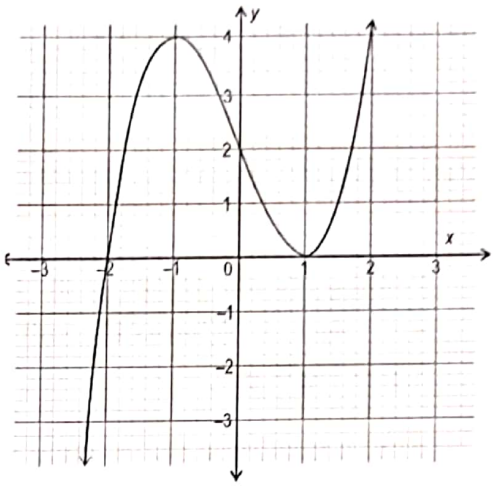
a)



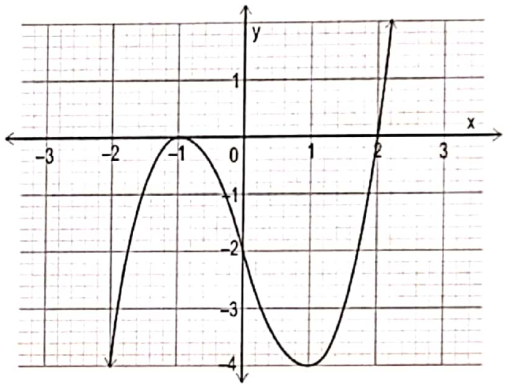
b)



c)



d)



4. A quadratic polynomial, the sum of whose zeroes is 0 and one zero is 4, is
   a) $x^2 - 16$ b) $x^2 + 16$
   c) $x^2 + 4$ d) $x^2 - 4$

5. The graph of a polynomial p(x) passes through the points (-5, 0), (0, -40), (8, 0) and (5, -30). Which among the following is a factor of p(x)?
   a) $(x - 5)$ b) $(x - 8)$
   c) $(x + 30)$ d) $(x + 40)$

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Very Short/Short Answer Type Questions

7. p and q are the zeroes of the polynomial $4y^2 - 4y + 1$. What is the value of $1/p + 1/q + pq$

8. Find a quadratic polynomial whose zeroes are $5 - 3\sqrt{2}$ and $5 + 3\sqrt{2}$.

9. If α and β are the zeroes of $x^2 - x - 2$, form a quadratic polynomial whose zeroes are $2\alpha + 1$ and $2\beta + 1$

10. If α and β are the zeroes of $f(x) = 2x^2 + 5x + k$ such that $\alpha^2 + \beta^2 + \alpha\beta = 21/4$, find the value of k.

11. If m and n are zeroes of the polynomial $(3x^2 - x - 2)$, find the values of the following without factorizing the polynomial.

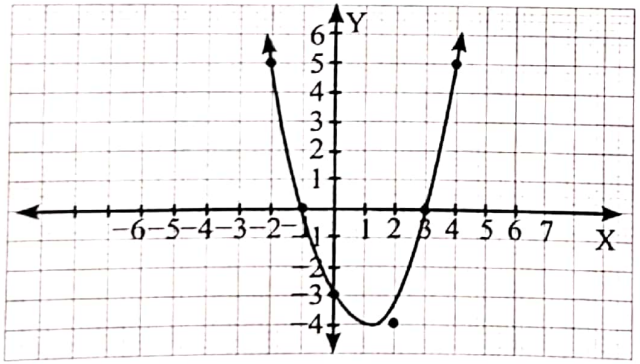
    i) $1/m + 1/n$

    ii) $m2 + n2$

    show your steps.

Case Study Based Questions

12. Due to heavy storm electric wire got bent as shown in the figure. It followed a mathematical shape. Answer the following questions below.



i) Name the shape in which the wire is bent

a) Spiral b) Elliptical
c) Linear d) Parabola

ii) How many zeroes are there for the shape the polynomial.

a) 2 b) 3
c) 1 d) 0

iii) What will be the expression of the polynomial?

a) $x^2 + 2x - 3$ b) $x^2 - 2x + 3$
c) $x^2 - 2x - 3$ d) $x^2 + 2x + 3$

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